Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Acceleration

Meaning in my words How would you explain acceleration to a friend?	**How can you tell?** A cart follows a bend at steady speed. Why is it accelerating?
Example or sketch Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits?	**Non-example and why** A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating?

Speeding up

Meaning in my words How would you explain speeding up to a friend?	**How can you tell?** A cart changes from −2 to −5 m/s. What happens to its speed?
Example or sketch Draw three equal-time dots for a cart speeding up to the right.	**Non-example and why** A cart moves right with acceleration to the left. Is it speeding up before it turns?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slowing down

Meaning in my words How would you explain slowing down to a friend?	**How can you tell?** Velocity changes from −5 to −2 m/s. What happens to the speed?
Example or sketch Draw velocity and acceleration arrows for a bike braking while moving right.	**Non-example and why** “Negative acceleration always means slowing down.” What happens if velocity is negative too?

Turning

Meaning in my words How would you explain turning to a friend?	**How can you tell?** A rider keeps the speedometer at 4 m/s around a bend. Which part of velocity changes?
Example or sketch How would you draw two velocity arrows as a cart turns from east to north at steady speed?	**Non-example and why** A cart brakes on a straight line and still moves east. Has it turned?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the before and after velocity arrows (m/s). Mark the change in velocity (m/s). Divide by time to label the acceleration (m/s²).

Before

After

My marks mean

On assignment: Acceleration

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two signed XJ velocities are −2 and −6 km/s, two seconds apart. The control room must choose its speed alert.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Telemetry engineer**

Send the average acceleration with its sign and unit. Sketch before and after velocity arrows so the shift lead can confirm the alert.

☐ **Replay commentator**

Narrate what changed over two seconds. Have your producer show which arrow proves whether XJ sped up or slowed down.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.